Oracle

Retrained on Homology-Restricted MegaScale

Zero-shot

Supervised, not trained on Megascale

Rectified gain @ k=96 (kcal/mol)

Predicted most stabilizing per domain

100
10
1
0

27.80 14.81 3.38
17.69 9.11 1.62
19.10 9.77 1.87
14.26 6.35 1.04
15.66 7.74 1.40
16.99 7.79 1.48
10.31 3.50 0.39
12.41 4.49 0.73
2.28 0.27 0.00
11.84 4.34 0.82
11.39 5.91 1.30

50.65 34.30 11.04
51.83 33.61 10.57
51.65 33.54 10.81
54.23 33.45 8.32
48.78 35.99 11.23
49.97 35.17 10.86
61.61 26.70 7.70
54.30 32.65 9.04
55.22 33.04 7.74
51.65 35.35 9.00
46.43 38.22 11.35

Top 96 per domain
ESM-MSR (p0) WT head
ESM-MSR (p0) combined
MutateEverything st1
ThermoMPNN
SPURS
ESM3-small-open unmasked
ProteinMPNN
ProSST
GeoDDG-3D
Rosetta

% conservation ≤ 25
25 < % conservation ≤ 75
% conservation > 75

Top 96 per Homology-Restricted MegaScale test domain (23 domains with the feature, 27,163 single mutants); gain = sum of max(ddG, 0) over the picks predicted stabilizing (> 0); boxes: interquartile range with darker colored median line; dots: domains; dark bars & annotations: mean over domains; dashed lines: oracle mean.